Does the books' advice hold at his level?

Nine rules from two chess books, tested against Stockfish and Maia-2 on Lichess 10-minute games and his own

Chess Coach research lab

Abstract

Two chess books for club players answered twenty questions the coaching app raised. Where an answer suggests a rule the app could follow or a number it could aim at, this paper tests it before the app acts on it, each under a rule written down and committed before its run, on 74,000 rated 10-minute Lichess games and his 639. After the opponent's serious error, 62% of replies at his level punish it; 17% of the failures take material that is a trap, enough for a drill that mixes gifts with traps, and the rest are misses whose best reply is rarely a fork, a mate or a pin, so the punish goal's practice stays on hanging pieces. Four of eight warning signs of danger to one's own king clear the bar at his level, but past the first 12 plies, where a warning in a game can speak, only two hold: the enemy queen on the king's exits and a king with no safe square (lifts 2.18 and 2.04); the rook lift and two more attackers than defenders mostly mark a later stage of the game. The coach's "good trade" praise is faulted on 2.8% (2.0–4.0) of praised trades at his level, and no case the books warn about makes it worse. Backward moves are not the moves his level misses most: a quiet backward best move is missed less often than a quiet forward one, and Maia-2 finds it more likely. In lost positions, Maia-2 is not a fair judge of which move sets the most problems, so the review keeps naming the engine's top move. The books' 70% conversion target is already passed by players at his level; the app's three conversion goals were above what players 200 points above each stage convert, and move down to those rates. On the books' twelve planning positions, the guess-the-plan drill's 45 cp rule agrees with every verdict at its own budget. Two later tests put the books' warnings about winning and losing positions to the same kind of rule. Serious errors are rarer, not more common, in the three moves after a player wins material than in positions alike, so the scan stops singling out that moment. When two pawns down and losing, an even trade of pieces saves 1.7 fewer games in 100 than keeping the pieces, under the bar, so the app says nothing about it.

1 The question

Levy Rozman's *How to Win at Chess* (2023) and *How to Win at Chess: Next Level* (2026) answered twenty questions about the app's coaching. Most answers are qualitative. Seven suggest a rule or a number, and each became a test (Table 1):

- **1a** After the opponent's serious error, how often do players at his level punish it, miss it, or take material that is a trap?
- **1b** Which warning signs of danger to one's own king does the engine bear out?
- **1c** How often does the coach's "good trade" praise land on a trade the books would not praise?
- **1d** Are retreats, and backward knight moves above all, the best moves his level misses most, and the hardest for Maia to find?
- **1e** When every move loses, does the move whose winning reply is hardest to find save more games than the engine's top move?

1f What share of decisive advantages do players at his level and 200 points above convert, against the app's goals?

1g Does the guess-the-plan drill's 45 cp rule agree with the books' verdicts on their twelve planning positions?

A second look at the answers gave two more:

6a Are serious errors more common in the three moves after a player wins material than in positions alike?

6b When down material and losing, does an even trade of pieces lower the chance of saving the game?

No model judges anything here. Every verdict comes from Stockfish, Maia-2 or the app's own board code, imported through the lab's bridge at app commit `50de20ccbf` (6a and 6b, run later, at `fa07ea1ed8`), and each test's rule, its bar and what the app does when it passes were committed before its run (the project's `README.md`).

test	what the rule found	what the app took
1a	Traps are a real share of the failures, and misses dominate	A "take it or not" drill (chess PR 109); the punish goal keeps its hanging-piece practice: no other kind reaches one miss in ten
1b	Four of eight signs warn; past the book, two	Practice games warn on their reply (chess PR 111)
1c	No case tightens the praise	Nothing: the praise stays
1d	Backward moves are missed less often, not more, and are not harder for Maia	Nothing
1e	Maia is not a fair judge here; the practical try does not save more	Nothing: the review names the top move
1f	All four conversion numbers move	The goals and the tripwire (chess PR 108)
1g	No disagreement holds	Nothing: the margin stays
6a	Serious errors are rarer, not more common, right after winning material	The scan drops its clause on winning material
6b	An even trade when lost costs fewer saved games than the bar	Nothing

Table 1: The nine tests, what each rule decided, and what the app took.

2 The data

The games are the 74,000 rated 10-minute Lichess games of September 2026 (CC0) that the think-time study keeps, with Lichess's evaluation on every move and the clock. Each gives two player-games, one for each player, read the app's way. A player-game's band is the player's own Lichess rating. His chess.com rapid rating comes to the middle of his band, 1250–1449, by the app's conversion; the band above is 1450–1649. His band holds 39,903 player-games and the band above 34,973. His games are his 639 10-minute games in the app's archive at that commit; only aggregates are committed. The engine is the app's Node engine service (Stockfish 18 lite, one thread a search), at the app's own budgets, and Maia-2 comes through the same service. The books' positions (test 1g) are not in this repository: the test reads them at run time and commits counts only. Tests 6a and 6b need no engine: they read the games' own evaluations and the app's board code, and at their commit his archive holds 647 10-minute games.

Shares carry 95% Wilson intervals.

3 1a. The move after their mistake

The books warn that a piece offered for free may be a trap, and that the usual process applies after the opponent's mistake: check what changed, look at the forcing moves, test the reply. The app's "Punish their mistakes" goal drills hanging-piece puzzles.

The rule. The event is a player's move right after the opponent's serious error (a mistake or a blunder by the review reader). The reply *punished* it when it kept at least half of the gift, the app's own test; it took a *trap* when it did not and was a capture the reader judged a mistake or a blunder; otherwise it *missed*. Traps are a real share when at least one failure in ten is a trap in his band, or in his games with the interval's lower end there or above; then the app gets a "take it or not" drill and the review tags a poisoned capture. Misses dominate when at least two failures in three are misses in his band; then the goal's practice adds the punishments it lacks, by what the misses missed.

The split. Both hold (Table 2). At his level 62.0% of replies punish the error. Of the failures, 17.1% (16.8–17.4) are a trap taken and 82.9% (82.5–83.2) a miss; his own games split the same way (18.9% (16.1–22.1) traps). On the punish goal's own population, gifts of 20 win% or more, traps are a larger share of the failures (21.8% (21.3–22.4) at his level). After a punished reply, the player's next move is a serious error 21.7% (21.5–22.0) of the time, against 9.6% (9.5–9.7) on his other moves at 75 win% or more: the books' "success is not permission to stop looking".

		share of the replies			share of the failures	
players	replies	punished	a trap taken	missed	a trap taken	missed
his band	144,404	62.0%	6.4%	31.1%	17.1% (16.8–17.4)	82.9% (82.5–83.2)
the band above	121,595	63.4%	6.0%	30.2%	16.7% (16.3–17.0)	83.3% (83.0–83.7)
his games	1,927	65.3%	6.4%	27.6%	18.9% (16.1–22.1)	81.1% (77.9–83.9)

Table 2: The reply to the opponent's serious error.

What the misses missed. The engine (600,000 nodes) found the best reply in a seeded sample of 2,000 misses in each band and in all 531 of his, and the app's labeller named it (Table 3). No kind of punishment beyond the hanging piece names one miss in ten in his band: a mate 0.8% (0.5–1.3), a fork 1.6% (1.1–2.2), a pin, skewer or discovered attack 5.7% (4.8–6.8). The punish goal keeps its hanging-piece practice. The free piece itself is the best reply in 10.3% (9.1–11.8) of the misses and saving a loose piece of the player's own in 19.6% (17.9–21.3); the labeller does not name the rest (61.9% (59.7–63.9)).

Offered material. In his band, 3,000 sampled replies came right after the opponent's move left one of their knights, bishops, rooks or queens newly hanging. Taking it with the cheapest attacker was a trap, a mistake or a blunder by the app's judge, in 12.6% (11.5–13.8) of them, most often for a knight, 17.5% (15.3–20.0), and least often for a queen, 3.2% (1.9–5.6). The players took 27.8% (23.5–32.5) of the traps and left 34.1% (32.3–35.9) of the real gifts, so leaving a gift was the commoner failure. In his games 13.6% (9.8–18.6) of 228 offers were traps.

the engine's best reply	his band	the band above	his games
misses searched	2,000	2,000	531
the hanging piece	10.3% (9.1–11.8)	9.5% (8.3–10.9)	10.9% (8.6–13.9)
a mate	0.8% (0.5–1.3)	0.6% (0.3–1.1)	1.5% (0.8–2.9)
a fork	1.6% (1.1–2.2)	1.7% (1.2–2.4)	2.5% (1.4–4.1)
a pin, skewer or discovered attack	5.7% (4.8–6.8)	5.0% (4.1–6.0)	3.8% (2.5–5.8)
the f2 or f7 strike	0.2% (0.1–0.5)	0.1% (0.0–0.4)	0.0% (0.0–0.7)
saving a loose piece of his own	19.6% (17.9–21.3)	16.4% (14.9–18.1)	19.4% (16.3–23.0)
a move it does not name	61.9% (59.7–63.9)	66.7% (64.6–68.7)	62.0% (57.8–66.0)

Table 3: The engine's best reply in the position the opponent's serious error left, where the player missed the punishment, as the app's labeller names it.

4 1b. Danger to his own king

The books list warning signs that come before a mate is on the board: a king with no safe squares, an enemy queen covering its exits, a bishop aimed into the shelter, an approaching knight, a rook lift, a hook pawn, a pawn storm when the kings stand on opposite wings, and at least two more effective attackers than defenders near the king (*plus two*).

The rule. The positions are one position before a player's move from each of 8,000 player-games in each band, past the book, not in check and not an endgame by the app's phase divider, and up to ten from each of his games. A position is *dangerous* when the engine finds a forced mate for the opponent with the opponent to move (120,000 nodes), or the game's own evals put the player 30 win% or more lower within six plies. Each sign is board code over the app's board kernel, over the king's zone (its square, the eight around it and the three squares two ranks in front). A sign becomes a warning when, in his band, it shows in at least 2% of positions, lifts the danger rate at least 1.5 times, and its precision's interval starts at 1.25 times the base rate or more.

The result. Four signs clear the bar (Table 4). The queen on the exits lifts the danger rate most. The opposite-side storm lifts it 1.55 times, but its interval starts at 15.1%, under the floor of 15.6%. No safe squares foretells mate threats (lift 5.1) and not swings (0.93).

	his band				above	his games
sign	shows	dangerous when shown	lift	clears	lift	lift
every position		12.4% (11.7–13.2)				
no safe squares	3.6%	20.7% (16.4–25.7)	1.66	yes	1.80	2.55
queen on the exits	6.2%	30.3% (26.4–34.5)	2.43	yes	2.43	3.15
bishop aimed in	47.3%	14.4% (13.4–15.6)	1.16	no	1.19	1.01
approaching knight	72.2%	12.1% (11.3–12.9)	0.97	no	0.94	0.96
rook lift	24.1%	18.8% (17.1–20.6)	1.51	yes	1.58	1.72
hook pawn	57.8%	10.8% (9.9–11.7)	0.87	no	0.83	0.86
opposite-side storm	3.6%	19.3% (15.1–24.3)	1.55	no	1.43	1.85
plus two	6.5%	21.1% (17.9–24.9)	1.70	yes	1.60	2.19

Table 4: The eight signs in his band's positions (base rate: mate threats 2.3%, swings 10.3%), with each sign's lift in the band above and in his games.

Where a warning can speak. A step taken after the result. Inside the first 12 plies the signs say nothing: any sign was dangerous 7.6% (4.2–13.4) of the time against a base of 6.0% (1,755 positions), since the uncastled king, boxed in by its own pieces, shows "no safe squares" from the first move. A practice game from the start can only tell the book is over by waiting out those plies, so the same

bar was applied again to the 6,245 positions after them (Table 5). Danger rises with the ply, from 11.1% in plies 13 to 20 to 18.7% from ply 41, and the rook lift and plus two mostly marked that rise: past the book their lifts are 1.31 and 1.49, and against the danger each ply bin predicts they lift it little. Only the queen on the exits and no safe squares clear the bar there, and they keep a lift of 1.5 or more against the ply alone in every group. This step could take signs away, never add one.

	his band, after ply 12				against the ply alone		
sign	shows	dangerous when shown	lift	clears	his band	above	his games
every position		14.3% (13.4–15.2)					
no safe squares	2.9%	29.1% (23.0–36.1)	2.04	yes	1.88	2.01	2.71
queen on the exits	7.6%	31.1% (27.1–35.4)	2.18	yes	1.95	1.79	2.26
rook lift	30.7%	18.8% (17.1–20.6)	1.31	no	1.17	1.14	1.16
plus two	8.3%	21.3% (18.0–25.1)	1.49	no	1.32	1.17	1.44

Table 5: The four signs that cleared the bar, on his band's positions after the first 12 plies. "Against the ply alone" is the danger observed over the danger expected from the base rate of each ply bin (13–20, 21–30, 31–40 and 41 and later).

In his games, after the first 12 plies, a warning on the two signs, said when a sign appears that the position before his last move did not show, would speak 0.46 times a game (none in 65% of games; 0.53 in his band's). Danger follows it 26.7% (20.5–33.9) of the time, against 9.7% (8.8–10.6) where no sign shows. On all four signs it would speak 1.61 times a game, followed by danger 16.4% (13.4–19.9) of the time.

5 1c. The coach's "good trade" praise

When the player is ahead, the live coach praises an even trade: "Good trade: knight for knight keeps you up a piece, and with fewer pieces left that lead counts for more." The books qualify the advice: do not give a more active piece for a less active one, nor a piece one's attack needs, nor the queens while attacking, and judge the ending the trade leads to.

The rule. Every move where the coach's trade rule fires in his games, and a sample of 1,500 in each band, is read as the praise gate reads it, at the coach's own budgets; the praised trades are the population. A case tightens the gate when it holds for at least one praised trade in twenty (5%) in his band or in his games, and the game's own eval faults its praised trades (a loss of 2 win% or more) at least as often as the others. A piece is *more active* when it has three more legal moves or more. A trade leads to a *worse ending* when the player's next move is in the endgame and the game's own eval faults the trade, so every trade in that case is faulted and only its share can decide.

The result. No case tightens the praise (Table 6). The coach praises 73% of the trades its rule finds at his level, and the game's eval faults 2.8% (2.0–4.0) of them (31 of 1,094); in his games 5.7% (2.2–13.8) (4 of 70). Giving the more active piece is common, 28% of the praised trades, but those are faulted less often than the rest (1.6% against 3.3%). Queens off while attacking is faulted more (11.1%), on 3.3% of praised trades, under the one in twenty the rule asks. The review's missed-trade line fires on 32 of his 840 faulted moves while ahead; the best move's trade gives the more active piece on 7 of 32, trades an attacker on 2 of 32 and takes the queens off while attacking on 2 of 32.

	his band			his games		
case	share	faulted	others faulted	share	faulted	others faulted
active for passive	28.1%	1.6%	3.3%	32.9%	1 of 23	3 of 47
an attacker traded	2.8%	0.0%	2.9%	4.3%	0 of 3	4 of 67
queens off while attacking	3.3%	11.1%	2.5%	2.9%	0 of 2	4 of 68
a worse ending	1.3%	all	1.6%	4.3%	all	1 of 67

Table 6: The books' cases among the coach's praised trades: each case's share, how often the game's own eval faults the praised trades in it, and the other praised trades.

6 1d. Backward moves

The books call backward knight moves among the hardest moves to notice, and ask the solver to check whether a retreat helps one or two moves later.

The rule. The positions are the position before a player's move, past the book, with at least two legal moves: up to ten from each of 3,000 player-games in each band, and 6,000 from his games. Only *must-find* positions count, where the engine (120,000 nodes, two lines) finds a best move at least 10 win% better than the second, so that missing it is at least a mistake by the app's judge. The player *found* it when he played the engine's best move. A best move is *backward* when it ends on a rank nearer the mover's own back rank, *forward* when nearer the opponent's, and *sideways* otherwise; castling is left out. The tests compare *quiet* best moves (no capture, no promotion), since a capture is seen as a capture whichever way it goes. Backward moves are *missed more often* when, in his band, the miss rate of backward quiet best moves exceeds that of forward ones with the difference's interval (Newcombe's) above zero. They are *harder for Maia at the same value* when Maia-2's probability for the best move, at the player's band against the opponent's, averages lower for backward than for forward quiet best moves within each of three bins of the best move's margin over the second (10 to 20, 20 to 40 and 40 or more win%), and the difference, weighted by the backward moves in each bin, has a bootstrap interval below zero. When both hold, the scan adds "look at backward moves too" and the review names a missed retreat.

The result. Neither half holds, and both point the other way (Table 7). In his band a quiet backward best move was missed 36.5% (30.7–42.7) of the time, against 46.7% (43.0–50.3) for a quiet forward one, a difference of −10.2 (−17.0 to −3.0) points. The band above is the same, and in his games the two are missed about equally. Backward knight moves were missed 32.8% (22.1–45.6) of the time against 45.0% (36.6–53.6) for forward knight moves, inside the noise. Maia-2 gives a backward best move more probability than a forward one at the same margin, not less (Table 8), in the two bins that hold 235 of the 244 backward moves; only in the last, with nine of them, does it give less. With captures counted, backward and forward best moves are missed about equally (30.8% (26.4–35.6) against 27.9% (26.1–29.7)), since only 31% of forward best moves are quiet against 63% of backward ones, and a capture is found more often. The move his level misses most is the quiet move forward. The scan and the review stay as they are.

		quiet best moves missed			
players	must-find	backward	sideways	forward	backward − forward
his band	3,101	36.5% (30.7–42.7)	37.8% (31.4–44.7)	46.7% (43.0–50.3)	−10.2 (−17.0 to −3.0)
the band above	3,050	36.3% (30.7–42.4)	40.4% (34.0–47.2)	47.2% (43.4–51.0)	−10.8 (−17.6 to −3.7)
his games	607	40.0% (27.6–53.8)	35.4% (23.4–49.6)	40.2% (31.9–49.0)	−0.2 (−15.4 to 15.9)

Table 7: How often the player missed a must-find best move, by the direction of a quiet best move. Differences are in points, with Newcombe's interval.

	backward		forward	
best move's margin (win%)	positions	Maia-2	positions	Maia-2
10–20	147	51.1%	466	43.8%
20–40	88	64.4%	206	57.0%
40 or more	9	61.5%	44	70.9%
weighted by the backward moves		backward − forward: 6.8 points (2.4 to 11.3)		

Table 8: Maia-2's mean probability for the best move in his band, by the best move's margin over the second, with the bootstrap interval of the weighted difference.

7 1e. Practical tries when every move loses

The books say that in a lost position the defender should make the opponent's task hard. Where every move leaves the player behind and he picked a worse one, the review says "You are behind either way" and names the engine's top move.

The rule. One position from each player-game where the game's own eval puts the player 250 cp or more behind (800 in each band, every one in his games), kept when the engine's top move still loses. The candidates are the engine's lines within 100 cp of the top move. For each, Maia-2 at the opponent's level gives the opponent's likeliest replies (up to six, until they cover 85%), and the engine scores them; a reply *throws the win away* when it leaves the player no longer behind. A candidate's *chance* is Maia's probability on those replies, and the *practical try* is the candidate with the highest chance. The practical try saves more games when Maia is a fair instrument (the replies that throw the win away rise from the lowest third of Maia's chance to the highest, and the predicted chance lies inside the observed share in at least two thirds), it differs from the top move in at least 25% of his band's positions, and its extra saves average at least 3 games per 100 there.

The result. It does not. Maia is not a fair instrument here (Table 9): in two thirds of the positions it gives no chance at all, yet some replies there do throw the win away; in the top third it is right. The practical try differs from the top move in 19.4% (16.6–22.7) of his band's 653 positions, under the bar. Where it differs, its extra saves would be 10.0 games per 100, from a mean chance of 18.0% against 14.0% for the top move. The game's own outcomes agree in direction: players who played the practical try were saved 46.7% (30.2–63.9) of the time (14 of 30), the top move 28.6% (23.0–34.9), any other move 19.2% (15.7–23.3). Where it differs, the practical try gives a check in 4% of positions, keeps pieces on in 12%, pushes a passed pawn in 2% and stalemates in 0%.

	his band		the band above		his games	
third of Maia's chance	observed	predicted	observed	predicted	observed	predicted
lowest	2.8% (1.3–5.9)	0.0%	1.0% (0.3–3.5)	0.0%	1.9% (0.5–6.8)	0.0%
middle	0.9% (0.3–3.3)	0.0%	1.0% (0.3–3.5)	0.0%	3.9% (1.5–9.6)	0.3%
highest	31.5% (25.7–38.0)	32.2%	32.4% (26.4–39.0)	32.3%	48.1% (38.7–57.6)	34.5%

Table 9: Maia's chance that the opponent's reply throws the win away, by third of the predicted chance, against the share of actual replies that did.

8 1f. The conversion target

The books' informal target is converting a decisive advantage at least 70% of the time. The app measures conversion as `conversion.sustained`: of the games where the player stood at +3 or better for three plies in a row, the share won. Its tripwire fires under 65% over the trailing 30 games, and its three stage goals ask for 78%, 82% and 85%.

The rule. A stage goal stays when it lies inside the interval of what players 200 points above that stage's band convert, and otherwise moves to that rate; the tripwire should fire when he converts no better than players 200 points below him.

The result. All four move (Table 10). Each goal was above what players 200 points above its stage convert, and the stage 4 goal asked for more than Lichess players at 1900–2049 convert, 77.9% (76.6–79.0). Players at his level already pass the books' 70%, converting 72.9% (72.4–73.5), and the band above converts 74.7% (74.1–75.3). In his 337 games with a sustained +3 he won 74.5% (69.6–78.8). His last 30 games at that commit hold 18 with a sustained +3, under the metric's minimum of 20, so the tripwire stays quiet until it has the games.

	chess.com	the band it reads (Lichess)	converts	was	now
stage 1 goal	1000–1150	1450–1599, 200 above	74.5% (73.9–75.2)	78%	75%
stage 3 goal	1300–1450	1750–1899, 200 above	75.9% (74.7–77.0)	82%	76%
stage 4 goal	1450–1600	1900–2049, 200 above	77.9% (76.6–79.0)	85%	78%
tripwire		1050–1249, 200 below him	71.4% (70.6–72.2)	65%	71%

Table 10: The conversion goals and the tripwire, the band each is read from, and what it moves to.

9 1g. The twelve planning positions

The guess-the-plan drill shows a quiet position with the engine's three best moves and counts any option within 45 cp of the best as right. Next Level's last chapter gives twelve positions with the author's preferred, sound and rejected moves: 5 rejected and 14 preferred or sound over eleven positions, and one that asks for an evaluation.

The rule. Each position is searched at the drill's own budget (600,000 nodes) and at 32 million nodes. A disagreement is a rejected move the rule counts as right, or a preferred or sound move it counts as wrong, and it holds when both budgets give it.

The result. None holds. At the drill's budget the rule counts zero rejected moves as right and zero preferred or sound moves as wrong; the deep check alone gives one and one. Of the rejected moves, one is among the drill's three options at all. The evaluation position, which the books call equal without saying who is to move, the engine reads at 32 million nodes as "about even" for White with

White to move and "much worse" for White with Black to move, so it disagrees. The margin stays.

10 6a. Errors right after winning material

The books warn that a tactical success is no licence to stop looking at the opponent: the author's own win of material against a strong player was followed by a forcing reply he overlooked. The app's scan ends: "Before you move: what is wrong with your move, and what is their best answer? Ask it hardest right after you win material."

The rule. A *capture run* is a stretch of consecutive plies that each capture or promote. A run *wins material* for a player when he captures or promotes in it and the material count (pawn 1, knight and bishop 3, rook 5, queen 9) stands two or more higher for him after it than before: a piece for a pawn, the exchange, a free piece or a new queen kept, but not a pawn or an even trade. The *window* is his next three moves after his last capture in the run, and the controls are his other moves, all past the book with a choice to make. A window move's expected chance of a serious error (the review reader's mistake or blunder) is the controls' rate in its stratum: his win% before the move in ten-point bins, tension (in check, right after a capture, or a piece hanging), either of the two signs of danger to his king that Play warns on, the phase, and his clock (under a minute, one to three minutes, three or more). It holds when, in his band, the window's serious errors exceed that expectation with a bootstrap interval above zero and their ratio, the *lift*, is at least 1.1. Then the routines' brief would get a reading for it; otherwise the scan drops its last sentence.

The result. It does not hold, and it points the other way (Table 11). In his band 37,151 runs won material, and 8.9% of the 94,543 window moves were serious errors, against 12.5% expected in positions alike: a lift of 0.71 (0.70–0.73). The band above and his games read the same way. Each of the three window moves is lower on its own (lifts 0.69, 0.70 and 0.75), and the more material won, the lower the lift (0.80 for 2, 0.70 for 3 or 4 and 0.56 for 5 or more pawns won). On the eval and the two dangers alone the lift is 0.78: the phase and the clock of the window moves, where errors are more common, widen the gap rather than close it. The moment after winning material is not where errors gather at his level, so the scan's last step drops "Ask it hardest right after you win material" and keeps its question for every move.

			serious errors			
players	runs	window moves	in the window	expected	lift	excess (points)
his band	37,151	94,543	8.9%	12.5%	0.71 (0.70–0.73)	−3.6 (−3.8 to −3.4)
the band above	28,081	71,410	8.4%	11.6%	0.73 (0.70–0.75)	−3.2 (−3.4 to −2.9)
his games	485	1,279	8.9%	11.3%	0.79 (0.63–0.97)	−2.3 (−4.4 to −0.3)

Table 11: Serious errors in the three moves after a run that wins material, against their expected number at the rate of other moves in positions alike. Lift and excess with bootstrap intervals over the player-games.

11 6b. Even trades when lost

The books' chapter on defending discourages even trades when down material, since fewer pieces leave fewer chances to resist, and allows the trade that improves the position, makes counterplay or makes a passed pawn. The app says nothing about trades when behind; it praises a trade only

when ahead (1c).

The rule. In each player-game, the first position where the player, past the book, not in check and after an opponent move that captured nothing, stands 300 cp or more behind by the game's own eval and two pawns or more down in material, with an even trade of pieces on offer: a capture of a knight, bishop, rook or queen whose exchange nets nothing, by the app's own trade reader. A *plain trade* takes it, without check and without making a passed pawn; *keeping the pieces* is a move that captures nothing; trades with check, trades that make a passed pawn, even pawn trades and other captures are counted apart. A game is *saved* when the player drew or won. Within each stratum (his win% before the move in five-point bins, the material down, his rating in 100-point bins, and whether the opponent is rated higher), the save rate after a plain trade minus after keeping the pieces, weighted by the plain trades. It holds when, over both bands together, that difference's bootstrap interval lies below zero and the drop is at least 3 saved games in 100. Then Play's coach and the review would say so where the engine agrees.

The result. It does not hold (Table 12). In 11,590 positions the players traded 33.1% (32.3–34.0) of the time. A plain trade saved 20.3% (18.9–21.7) of its games and keeping the pieces 22.2% (21.3–23.2); in positions alike the difference is −1.7 (−3.4 to 0.0) points, under the bar, with an interval that reaches zero. The trades themselves gave away less, 1.4 win% against 2.6 for the moves that keep the pieces, so at the same eval after the move the trade costs more: −3.4 (−5.2 to −1.7) points, reported without a rule. With the clocks as a further stratum it is −2.3 (−4.1 to −0.5). Trades with check save about as often as keeping the pieces, at −1.7 (−4.7 to 1.4) points, the 90 even pawn trades saved fewer, at −8.4 (−15.1 to −0.9) points, and the 42 trades that made a passed pawn are too few to read. His games hold 75 such positions, too few to read. Nothing changes.

			saved after		plain trade − keeping, in points	
players	positions	traded	a plain trade	keeping the pieces	alike before	alike after
both bands	11,590	33%	20.3%	22.2%	−1.7 (−3.4 to 0.0)	−3.4 (−5.2 to −1.7)
his band	6,370	33%	21.5%	23.6%	−1.9 (−4.4 to 0.5)	−3.5 (−6.0 to −1.1)
the band above	5,220	33%	18.7%	20.5%	−1.4 (−4.0 to 1.1)	−3.4 (−5.9 to −0.9)
his games	75	37%	21.7%	28.6%	2.5 (−27.8 to 29.4)	5.1 (−21.3 to 31.7)

Table 12: At the first position where a player two pawns or more down and losing could trade pieces evenly: how often he traded, how often the game was saved, and the difference in positions alike by the eval before the move (the rule) and after it.

12 Limits

The bands are other players, in games against their own level: their numbers say where the books' rules hold at his level, not what he would gain by following them. Danger in 1b is read from Lichess's evaluations six plies on and a mate search at a modest budget, so it misses slow attacks. The step after 1b's result applied a bar fixed before the run to a population chosen after it; it only took signs away. In 1a the labeller names one theme per best reply, has no trapped piece, and calls what it cannot name by the phase. Maia-2's probabilities in 1e cover the likeliest replies only, which is where it fails the test. In 6a the strata match what the app can read; a move right after winning material may still be easier in ways they do not hold. In 6b the trade reader judges a capture by its square, so six of the 3,836 trades lost material as played, and the cost at the same eval after the move would need its own rule and fresh games before the app acts on it. His Lichess level is the

app's conversion of his chess.com rating, and his games come from the app's reviews, with another engine and depth.